

DATE : 17.12.2024
DT/NT : DT
LESSON : LINUX
SUBJECT: LINUX ESSENTIALS
SESSION : 1
BATCH : B 303

AWS-DEVOPS



TECHPRO
EDUCATION



techproeducation.com



+1 (585) 304 29 59





Introduction to Linux

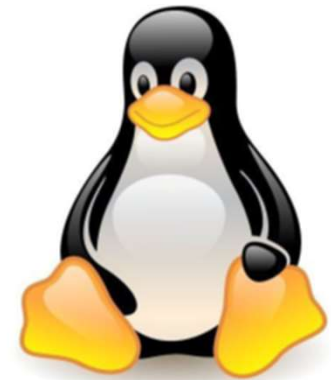


Table of Contents

- ▶ Why/Where/When Linux
- ▶ Linux Evolution
- ▶ Major Open Source Applications
- ▶ FSF and OSI
- ▶ Using Linux on Different Platforms
- ▶ What is SHELL?
- ▶ Case Sensitivity
- ▶ Simple Globbing
- ▶ Basic SHELL Commands-1



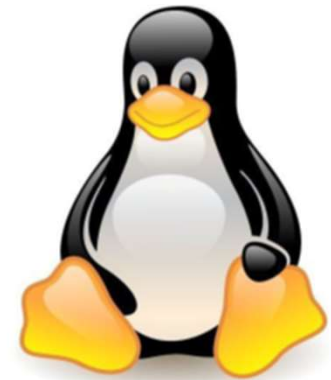


Why Linux?



Where Linux?

When Linux?



Why Linux?



- ★ Total cost of ownership
- ★ Beginner friendly and easy to use
- ★ Reliability
- ★ Hardware
- ★ Software

- ★ Security
- ★ Freedom
- ★ Annoying crashes and reboots
- ★ Server segment
- ★ Linux is everywhere

Where Linux?



Google™

<http://www.tecmint.com>

IBM
twitter



WIKIPEDIA

amazon.com™



London
Stock
Exchange

NYSE
New York Stock Exchange®

UnionBank™

30 Companies and Devices
Running on GNU/Linux

Where Linux?



1. Google

The services of which includes search, cloud computing and online advertising technologies runs on Linux.

2. Twitter

Twitter, famous online social networking and micro-blogging site is Powered by Linux.

3. Facebook

Facebook, one of the most famous and most widely used Social Networking service runs on the same platform.

4. Amazon

An American based international company which deals with International Online Retailing is in the list of Linux powered Company.

5. IBM

IBM (International Business Machine Corporation) the American based company which for sure don't requires any introduction, is again powered by Linux.

Where Linux?



6. McDonalds

The world's largest chain of hamburger fast food restaurant uses GNU/Linux (Ubuntu) too.

7. Submarines

The submarines in the United State Navy are controlled by same platform.

8. NASA

National Aeronautical and Space Administration, The United Nations Space program widely uses Linux in many of their programmes.

9. Watches

Most of you would not be knowing that there are Linux Powered Watches in the market, already. The watch developed by IBM running Linux.

10. Mobile Devices

True, you all know that Linux is powering Mobile Phones, Tablets and Kindle.

Where Linux?



11. Space

A Specific Linux Distro (Debian) is already in the space. Debian led all the rest.

12. Raspberry pi

The business card sized computer designed for electronic projects as well as desktop computing which is very cheap in cost and is fully functional. Raspberry is a landmark in Linux Development.

13. Desktop Computing

Though a little late, Linux made a notable presence in the desktop computing market. In school and academics as well as in government offices Linux are being widely used, these days.

14. Corporates

The corporate offices are using Linux and finds it more productive than any other alternatives.

Where Linux?



15. New York Stock Exchange

New York Stock Exchange (NYSE) which provides means for buyers and sellers in order to trade shares of stock in companies registered for public trading relies solely on Linux.

16. Traffic Controlling

The Traffic controlling system in most of the countries be it Road Traffic or Air Traffic Linux proved to be the best than any other available alternative.

17. Nuclear Projects

When it comes to Nuclear Ambitious projects, Linux is the best option. One of such OS is QNX, which lately is acquired by Blackberry Ltd.

18. Bullet Trains

The Bullet Trains in Japan runs at the speed of 240-320 km/h. All train tracking, maintenance, scheduling and controlling is Linux based.

Where Linux?



19. Internet Hosting

More than 70% of Internet Hosting and service providers are Linux based. Thought this statistic is difficult to figure out but based upon the Linux compatible hardware sold, and demand for cross platform compatible hardware, the above statistics is a rough estimation.

20. Missiles and Weapons

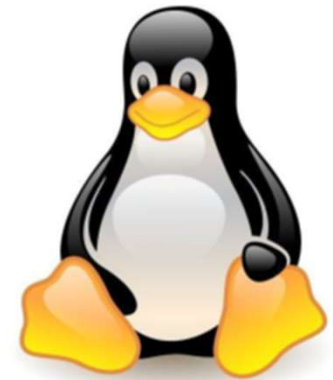
The Missiles and destructive weapons of next generation is themed to be much advanced and Intelligent system than its predecessors. Well what else would have been its alternative.

When Linux?



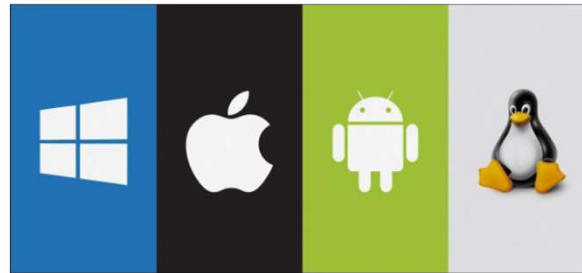


Linux Evolution



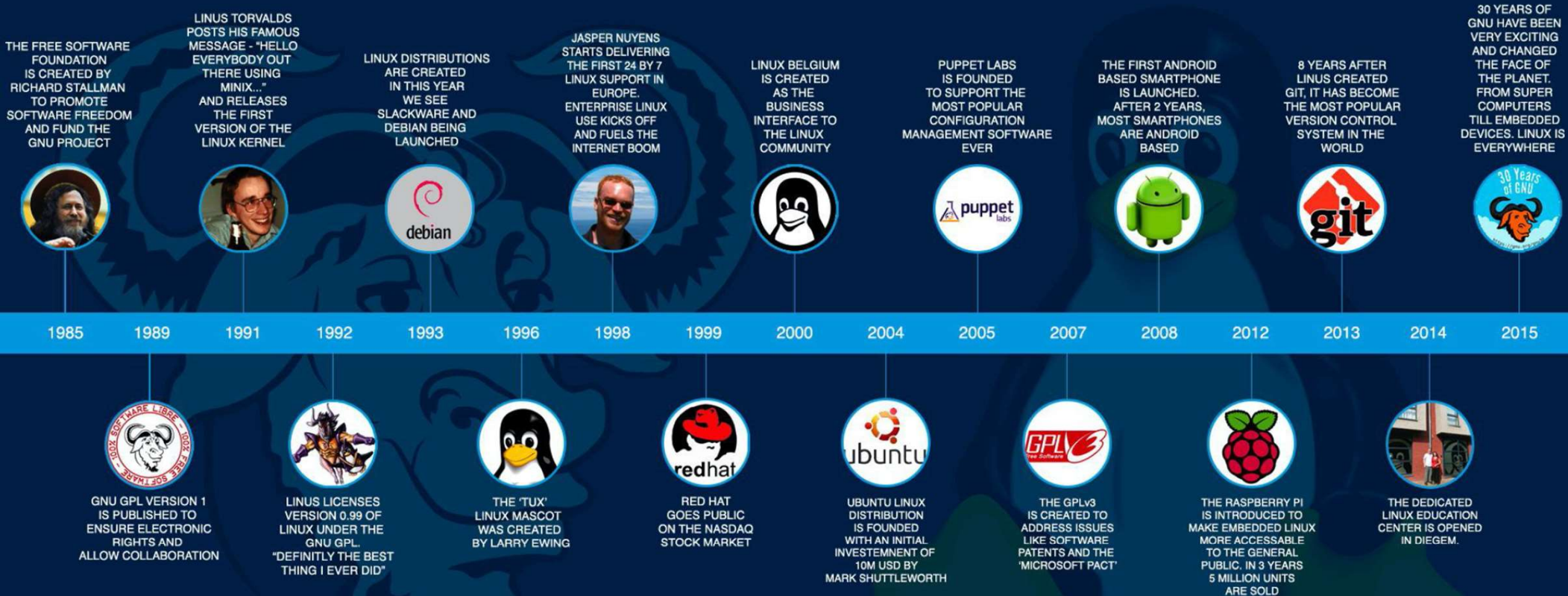
What is Linux?

- Free
- Open-Source
- OS



MEMORABLE LINUX EVENTS

CELEBRATING 30 YEARS OF GNU LINUX



Memorable Linux Events

1991



Linus Benedict Torvalds



Hello everybody out there using minix -

I'm doing a (free) operating system (just a hobby, won't be big and professional like gnu) for 386(486) AT clones. This has been brewing since april, and is starting to get ready. I'd like any feedback on things people like/dislike in minix, as my OS resembles it somewhat (same physical layout of the file-system (due to practical reasons) among other things).

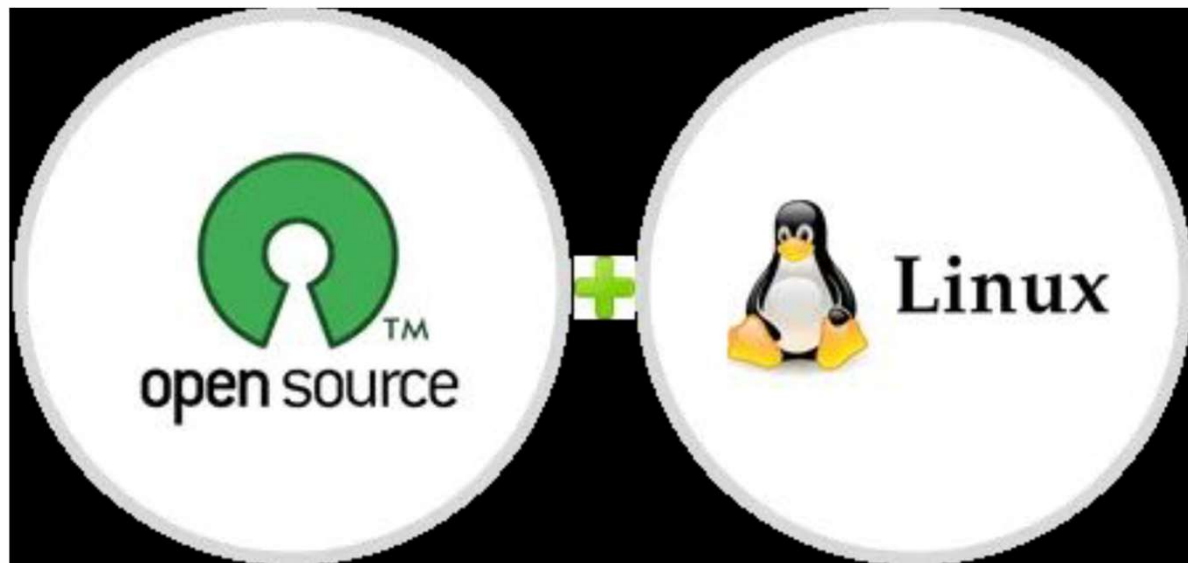
I've currently ported bash(1.08) and gcc(1.40), and things seem to work. This implies that I'll get something practical within a few months, and I'd like to know what features most people would want. Any suggestions are welcome, but I won't promise I'll implement them :-)

Linus (torv...@kruuna.helsinki.fi)

PS. Yes - it's free of any minix code, and it has a multi-threaded fs. It is NOT protable (uses 386 task switching etc), and it probably never will support anything other than AT-harddisks, as that's all I have :-).

Memorable Linux Events

1992

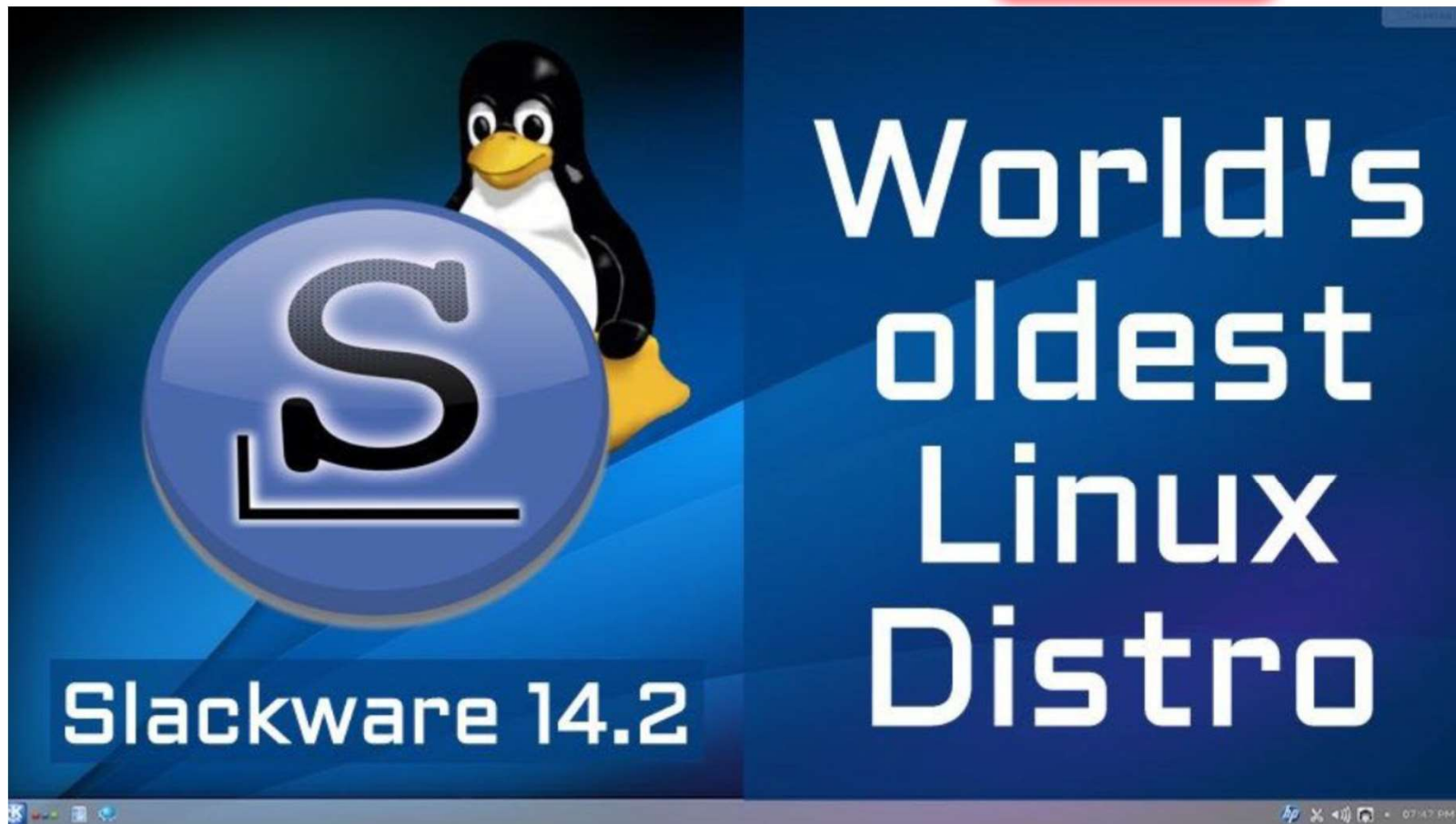


"Making Linux GPLed was definitely the best thing I ever did."

Torvalds, L.

Memorable Linux Events

1993



Memorable Linux Events

1995



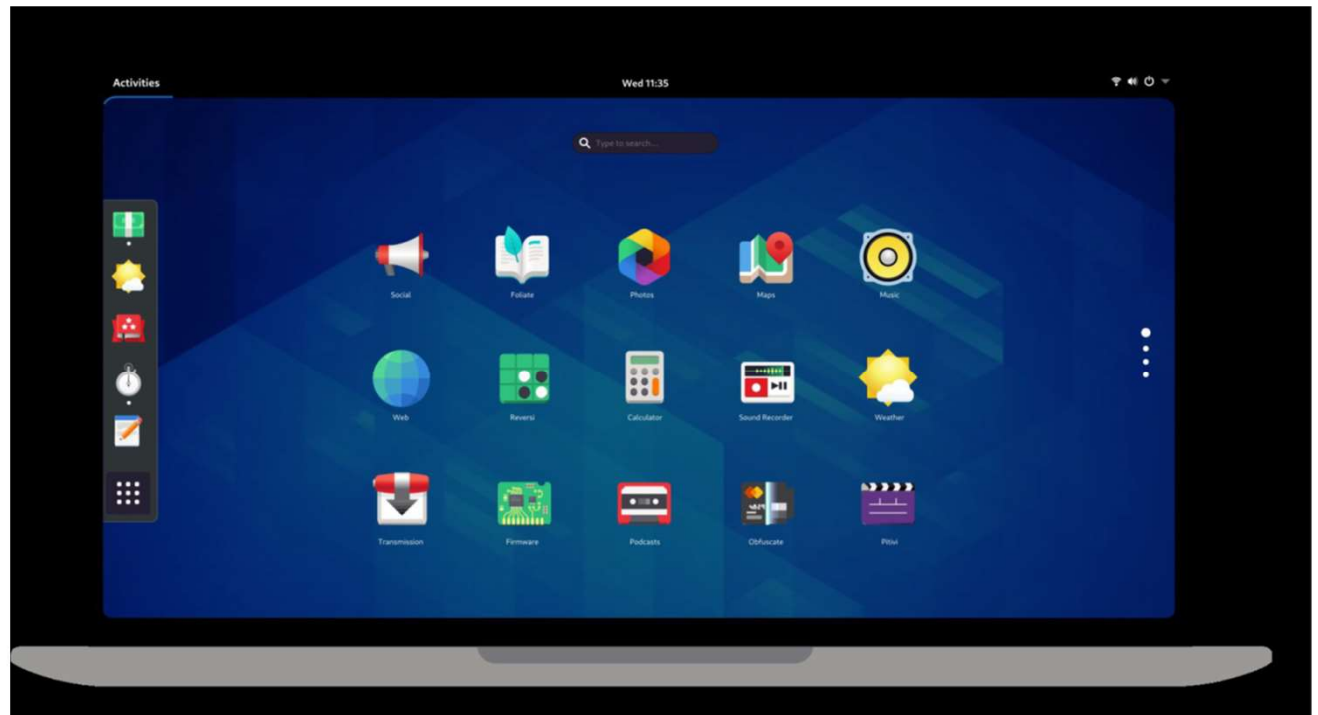
Memorable Linux Events

1996



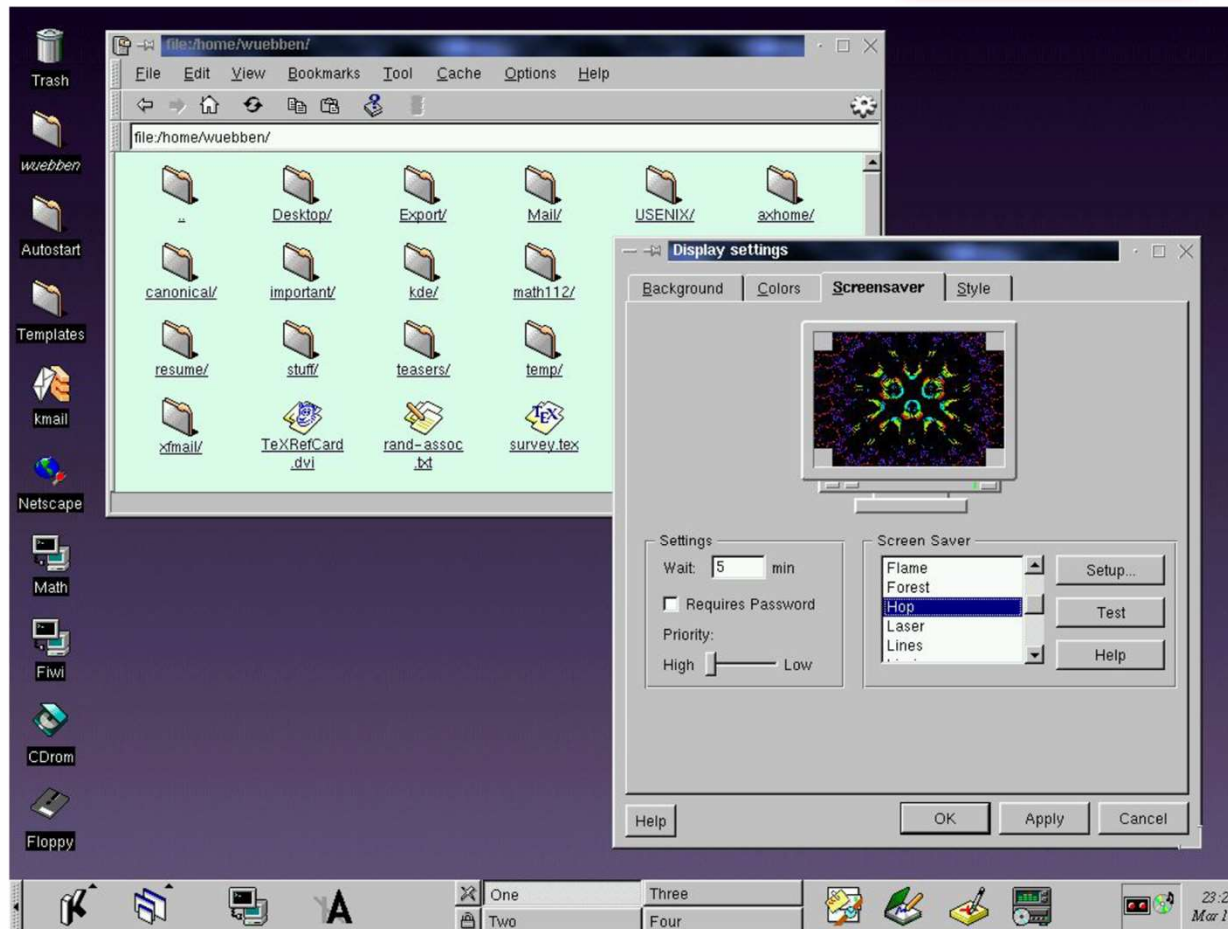
Memorable Linux Events

1997



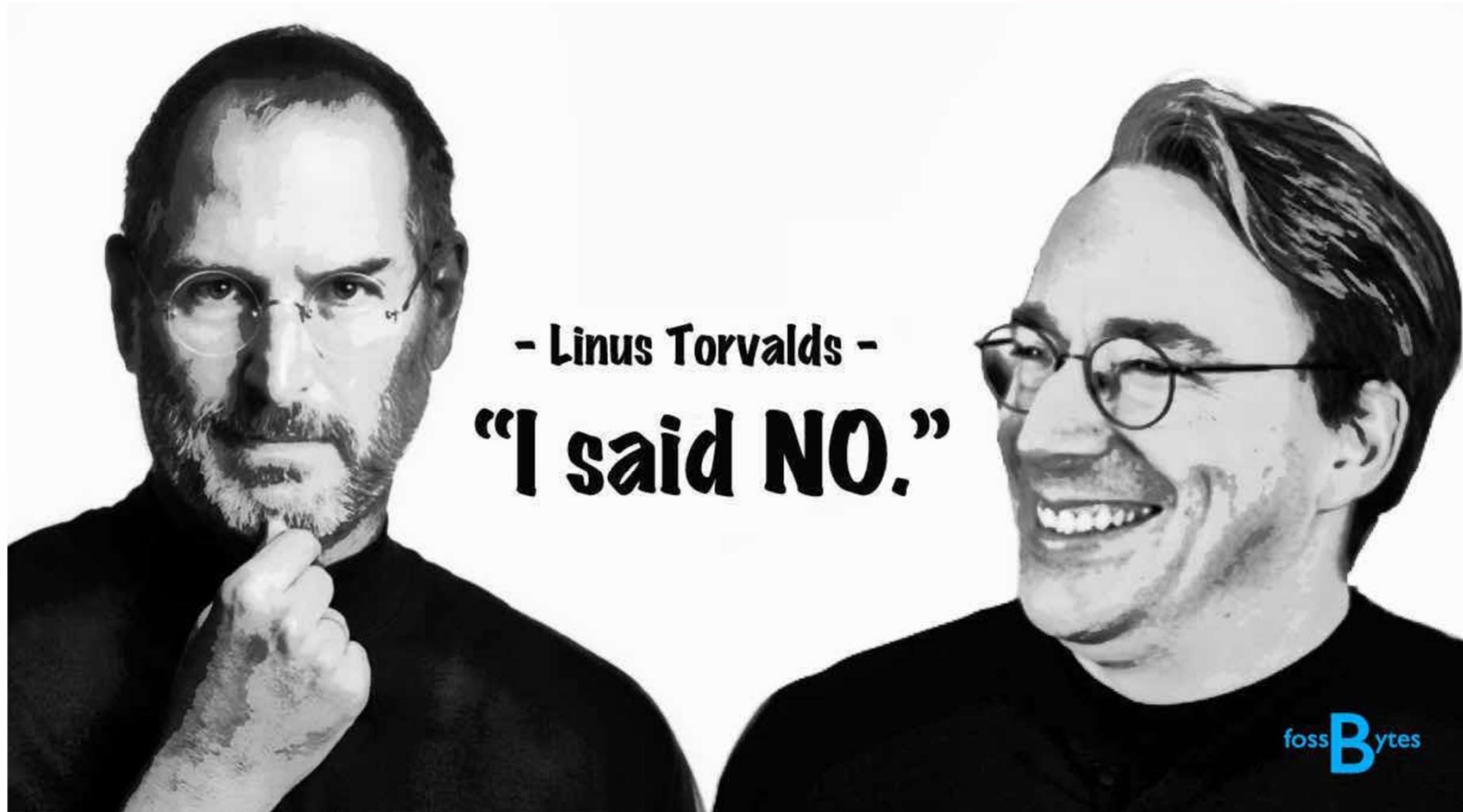
Memorable Linux Events

1998



Memorable Linux Events

2000



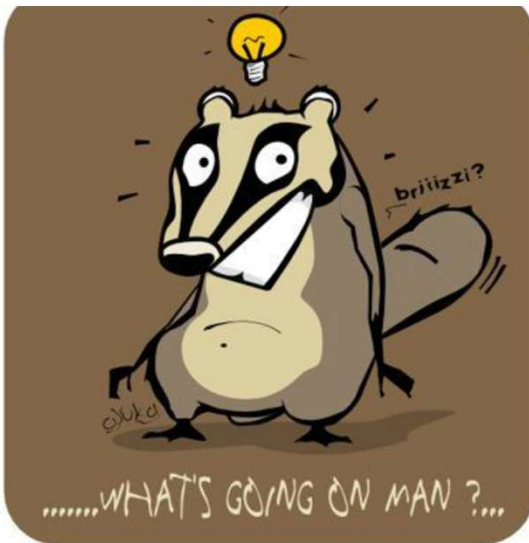
Memorable Linux Events

2002

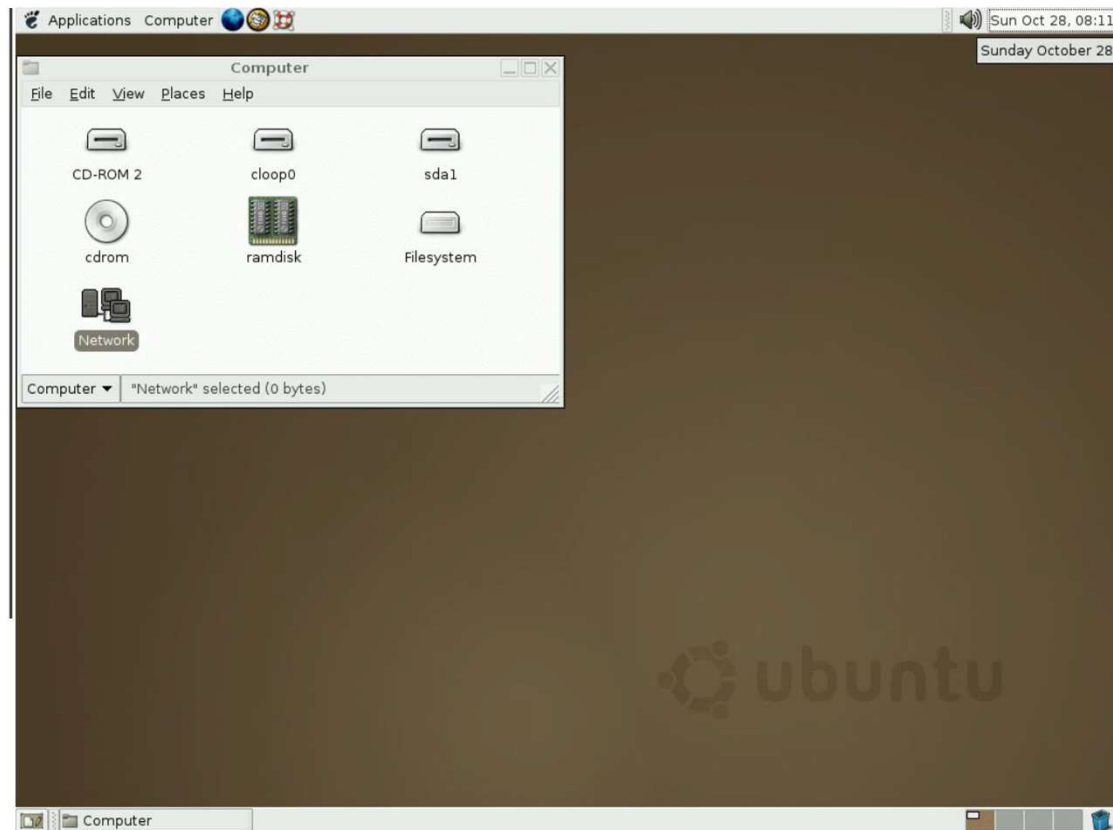


Memorable Linux Events

2004



 ubuntu



Memorable Linux Events

2005



Memorable Linux Events

2008



Memorable Linux Events

2009



Google™



Operating System

Memorable Linux Events

2014



“Microsoft loves Linux”
-Satya Nadella, 2014

Microsoft ❤️ Linux

Memorable Linux Events

2019



WSL 2 BRINGS

Linux Kernel in Windows 10

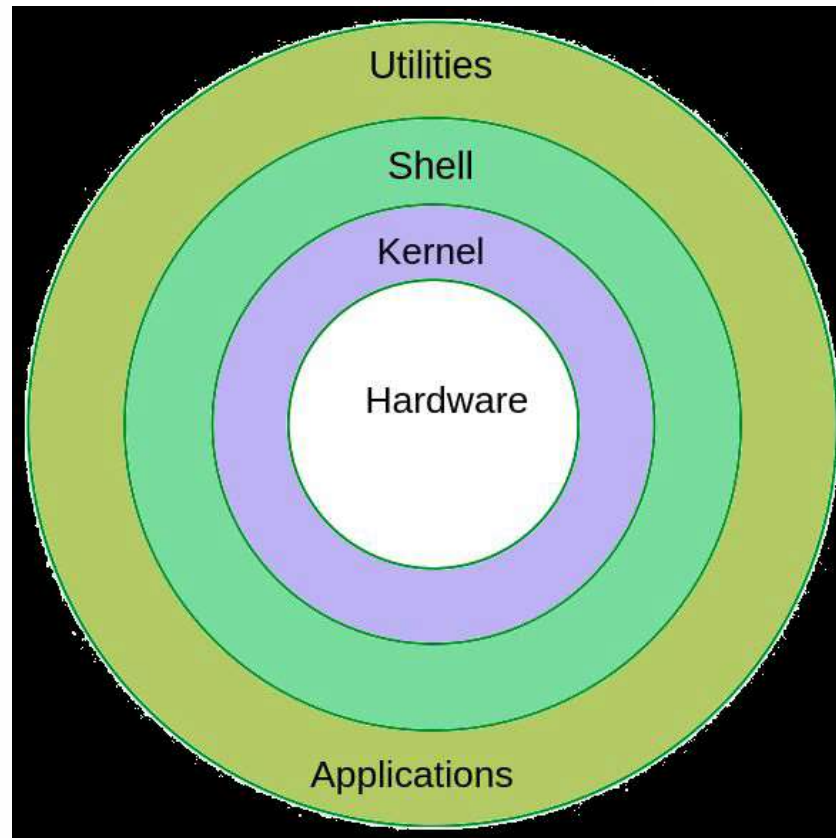
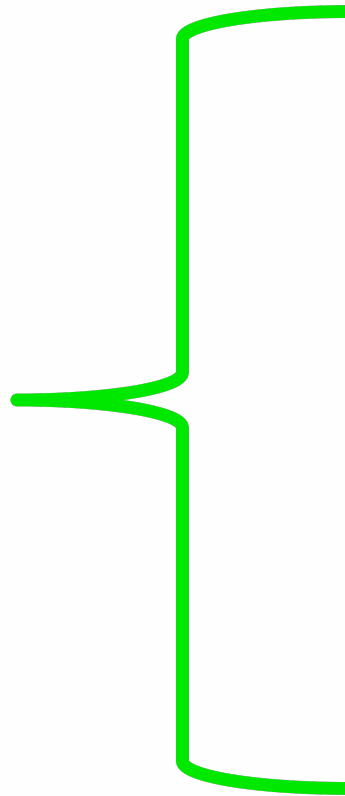
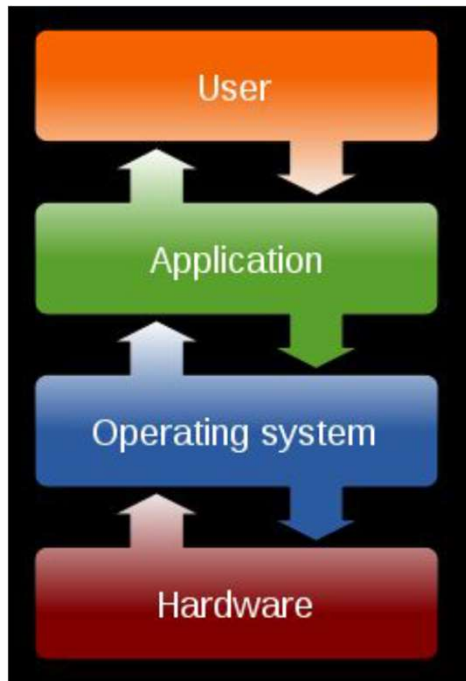
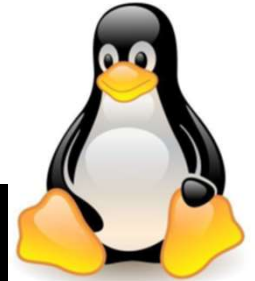
For Real!



WINDOWS SUBSYSTEM FOR LINUX
WSL 2 IS FAST!



Components of Linux

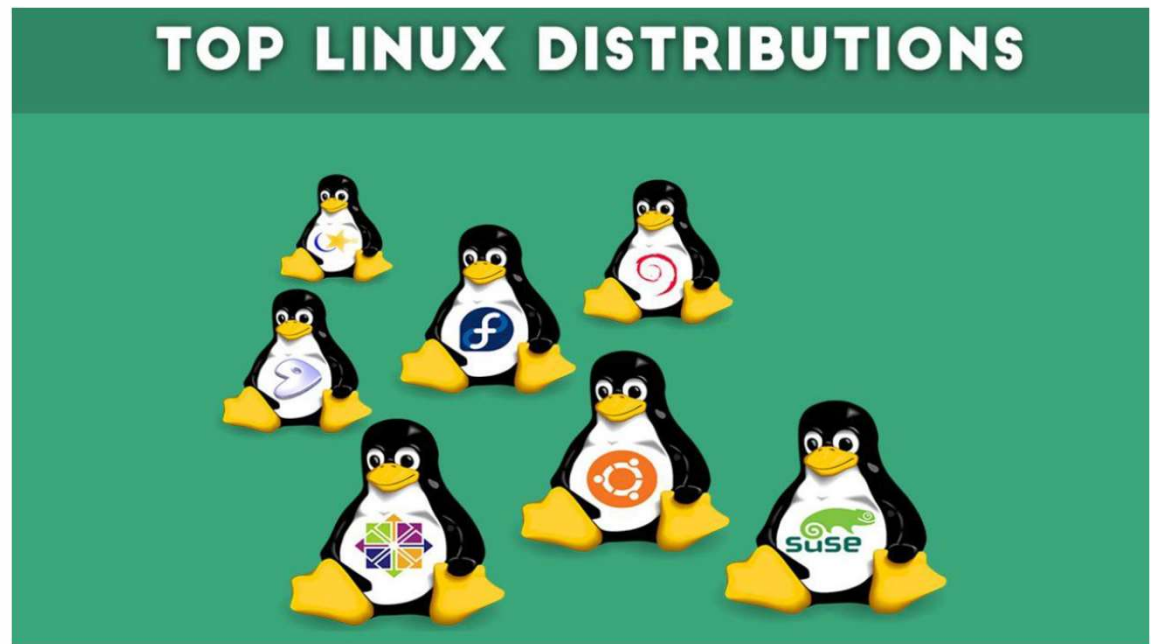


USER

Popular Linux Distributions



- Debian
- Ubuntu
- Centos
- Mint
- Manjaro
- openSUSE
- RedHat
- Fedora



Linux Embedded Systems



Embedded System

An embedded system is a computer system that is dedicated to one or two specific functions.



Major Open Source Applications



What is open-source?



Open-source software is software with source code that anyone can inspect, modify, and enhance.



Desktop Applications

BEST OPEN SOURCE SOFTWARE



1. LibreOffice
2. VLC Media Player
3. GIMP
4. Shotcut
5. Brave
6. Audacity
7. KeePass
8. Thunderbird
9. FileZilla
10. Linux



Server Applications



- Apache Web Server
- NGINX
- MySQL
- Samba
- ownCloud



Package Management Tools



Contemporary distributions of Linux-based operating systems install software in pre-compiled packages, which are archives that contain binaries of software, configuration files, and information about dependencies.

- **dpkg**: Debian Package Manager
- apt-get
- **rpm**: Red Hat Package Manager
- yum: yellowdog updater modified





**Free Software Foundation
(FSF)**

**Open Software Initiative
(OSI)**



FSF and OSI



Free Software Foundation (FSF)

- The Free Software Foundation (FSF) is a nonprofit organization with a worldwide mission to promote computer user freedom.
- The FSF is working to secure freedom for computer users by promoting the development and use of free software and documentation.



FSF and OSI



Open Source Initiative (OSI)

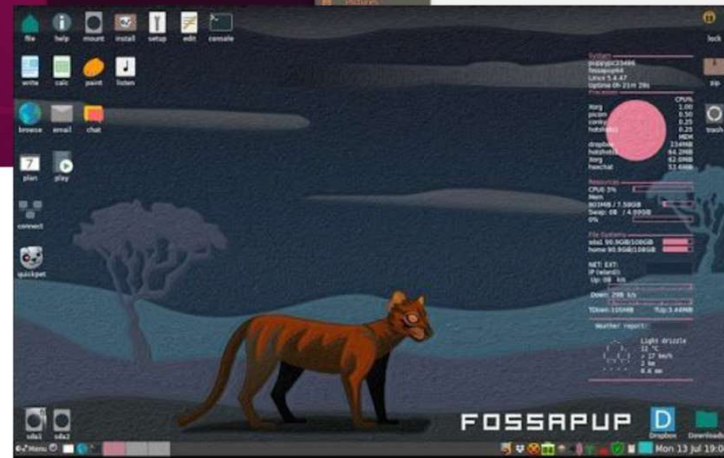
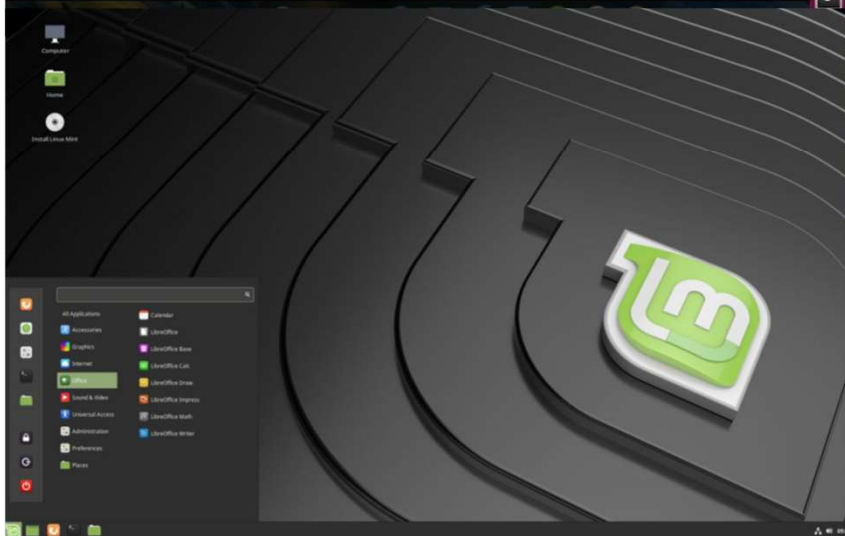
- The Open Source Initiative (OSI) is a non-profit organization dedicated to the promotion of open-source software.
- OSI was founded in 1998 by Bruce Perens and Eric Raymond.





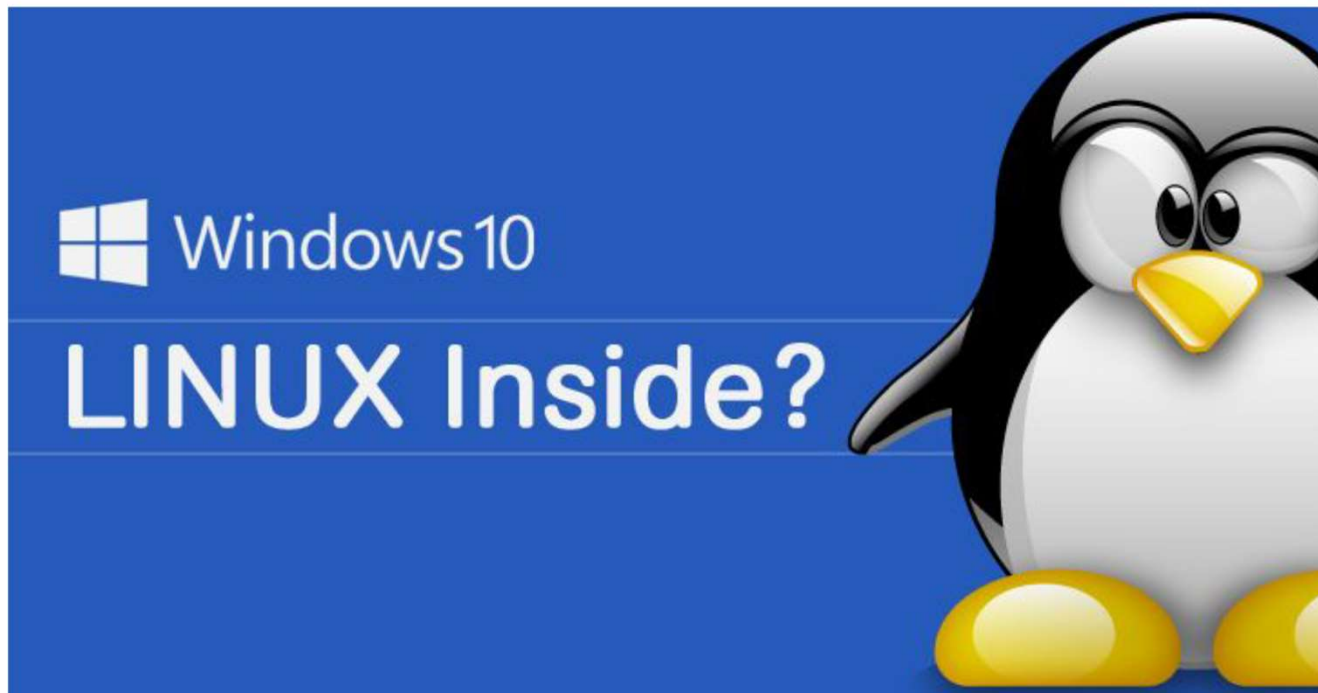
Using Linux on Different Platforms

Linux Alternatives



Linux Alternatives

WSL 2



Linux Alternatives

Linux Distros on Virtual Machines

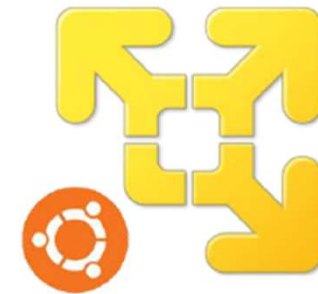


MacOS/Windows



ORACLE[®]
VM
VirtualBox

Windows



VMware Player

Linux Alternatives

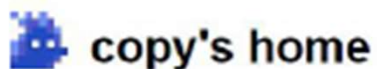
Linux Distro on Virtual Machines



Linux Alternatives



<https://copy.sh/>



You can email me at copy@copy.sh. Use my [GnuPG key](#).

Projects



Virtual x86

Run KolibriOS, Linux or Windows 98 in your browser.

linuxzoo.net



Welcome to linuxzoo

Learn Linux from the safety of your chair using a remote private linux machine with root access.

- Welcome to linuxzoo
 - [Our environment](#)
 - [Essential Linux](#)
 - [System Administration](#)

Status: Normal. Everything should be stable.

Look at the [Our Environment](#) link, and then [Running Your Machine](#) for getting started.

Quick start hints: register/login, Join Queue, Switch On (in Control tab), Wait for successful boot, click the Connect tab, and then click "telnet: linuxzoo.net" (or type `telnet linuxzoo.net` at your command prompt). Username `root`, password `secure`.

Image	Username	Password
Linux Centos 7	root	secure
	alice	secure
Caine Forensics 10.0	caine	caine

FAQ for VNC: There are a few options to getting a remote graphical desktop. In "connect" you can click on Java VNC, which requires java 7 installed on your machine. JavaScript VNC is more flexible, but it may be slower (it is experimental). Some systems do not like you logging in graphically as root.

Minimize

Login Details

Email:

Password:

Login | Register for an account

NOT currently queued

0 user(s) ahead in queue.

Refresh in: 2:00

Queue

Stats

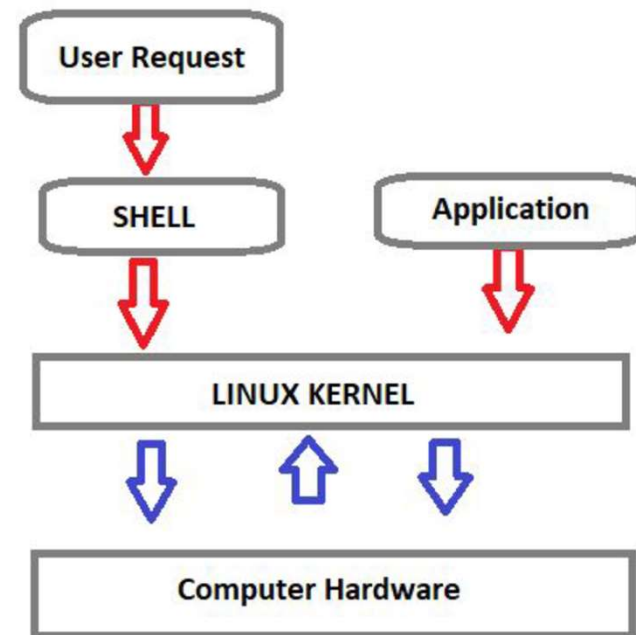
You must register and login before you can use a virtual machine



What is SHELL?

What is SHELL?

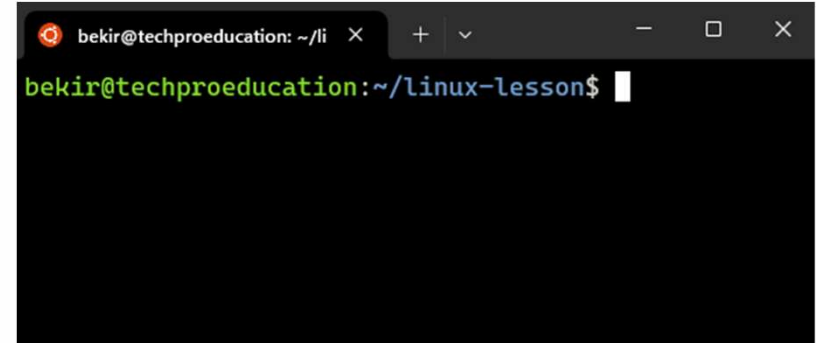
- Shell is a program that receives the user's commands and gives them to the operating system to process and displays the output.
- Bash (Bourne Again SHell) is an enhanced version of Steve Bourne's first Unix Shell application, and serves as the shell program on most Linux. systems



What is SHELL?



- The standard Linux shell is both a command-line interpreter and a programming language.
- The command prompt for Linux generally shows the current user, the current host, and the appropriate directory.
- At the end of the prompt list, the \$(dollar sign) signifies the current user being unprivileged, and the device is ready to receive feedback.
- The input is sent for parsing and execution to the interpreter.



Command Prompt



User Name

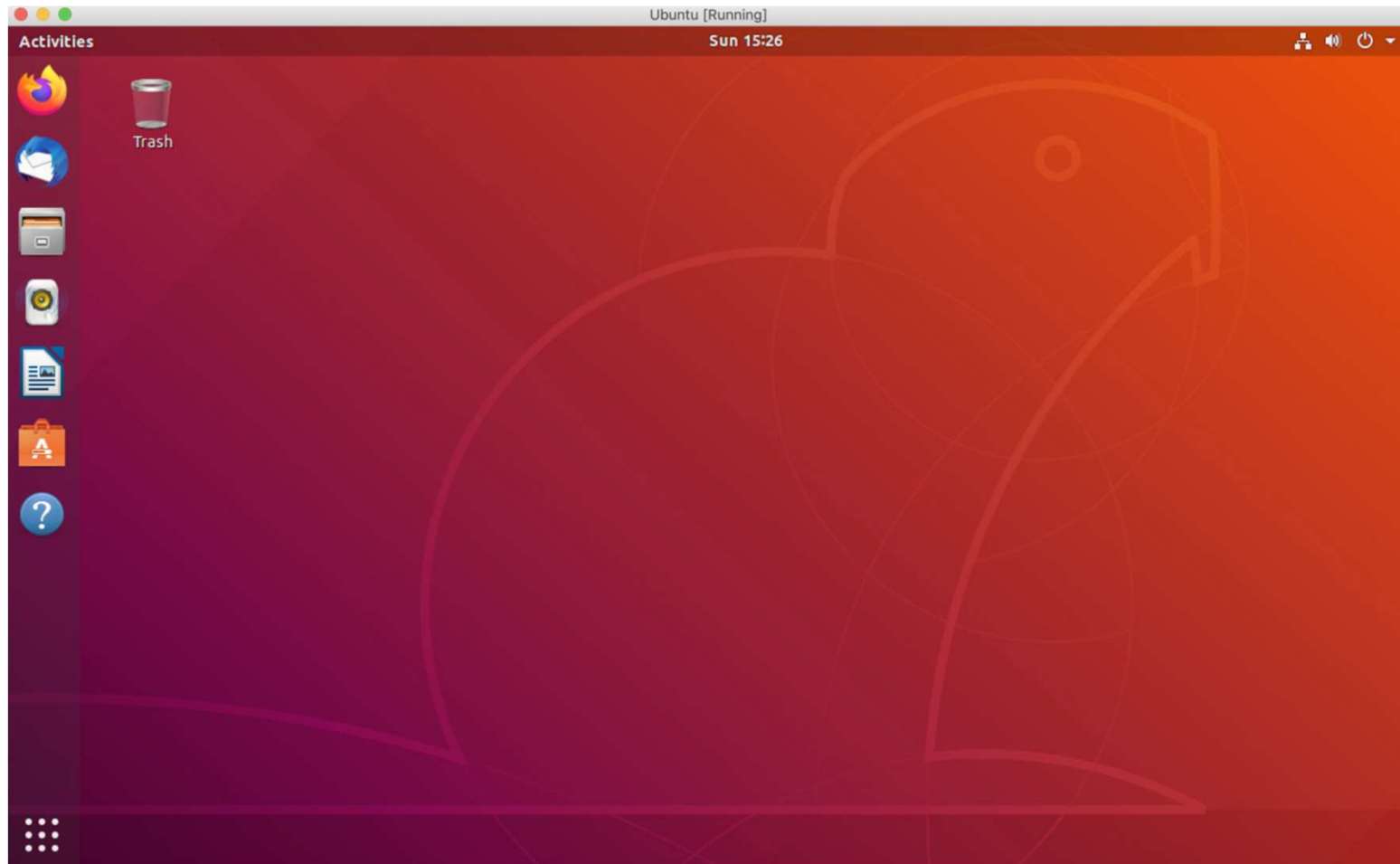
Host Name

Current Directory

A screenshot of a Linux terminal window. The title bar shows 'bekir@techproeducation: ~/li' and standard window controls. The terminal content shows the prompt 'bekir@techproeducation: ~/linux-lesson\$' with a cursor at the end. Three green arrows point from labels above to parts of the prompt: 'User Name' points to 'bekir', 'Host Name' points to 'techproeducation', and 'Current Directory' points to '~/linux-lesson\$'.

User Type
\$ normal user
Privileged user

GUI



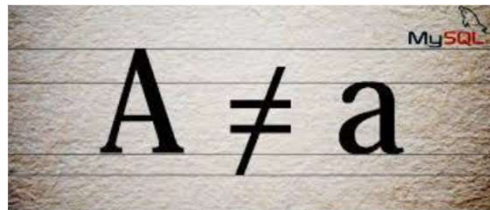


Case Sensitivity

Case Sensitivity



Most of the common Linux file systems are case sensitive; this is something to keep in mind when creating files or directories and moving through directories.



Lower-case and upper-case letters have different ASCII representation.

`touch newfile`
`touch Newfile`
Will create two different files.



Basic SHELL Commands-1

Basic Shell Commands



pwd show current path

```
bekir@techproeducation: ~/li x + v - □ X
bekir@techproeducation:/$ pwd
/
bekir@techproeducation:/$ cd
bekir@techproeducation:~$ pwd
/home/bekir
bekir@techproeducation:~$ cd linux-lesson/
bekir@techproeducation:~/linux-lesson$ pwd
/home/bekir/linux-lesson
bekir@techproeducation:~/linux-lesson$
```

Basic Shell Commands



- ls** lists directory contents
- ls -l** lists directory contents with details
- ls -al** lists all the contents with hidden ones

```
bekir@techproeducation: /  
bekir@techproeducation:/$ ls  
bin  dev  home  lib  lib64  lost+found  mnt  proc  run  snap  sys  usr  
boot  etc  init  lib32  libx32  media  opt  root  sbin  srv  tmp  var  
bekir@techproeducation:/$ ls -l  
total 2000  
lrwxrwxrwx 1 root root 7 May 2 2023 bin -> usr/bin  
drwxr-xr-x 2 root root 4096 Apr 18 2022 boot  
drwxr-xr-x 16 root root 3560 Nov 29 13:37 dev  
drwxr-xr-x 78 root root 4096 Nov 29 13:48 etc  
drwxr-xr-x 4 root root 4096 Oct 1 20:37 home  
-rwxrwxrwx 1 root root 1978872 Apr 20 2023 init  
lrwxrwxrwx 1 root root 7 May 2 2023 lib -> usr/lib  
lrwxrwxrwx 1 root root 9 May 2 2023 lib32 -> usr/lib32  
lrwxrwxrwx 1 root root 9 May 2 2023 lib64 -> usr/lib64  
lrwxrwxrwx 1 root root 10 May 2 2023 libx32 -> usr/libx32  
drwx----- 2 root root 16384 Sep 22 20:38 lost+found  
drwxr-xr-x 2 root root 4096 May 2 2023 media  
drwxr-xr-x 6 root root 4096 Sep 22 20:38 mnt  
drwxr-xr-x 3 root root 4096 Oct 1 21:28 opt  
dr-xr-xr-x 186 root root 0 Nov 29 13:37 proc  
drwx----- 4 root root 4096 Nov 25 21:19 root  
drwxr-xr-x 23 root root 680 Nov 29 13:58 run  
lrwxrwxrwx 1 root root 8 May 2 2023 sbin -> usr/sbin  
drwxr-xr-x 8 root root 4096 May 2 2023 snap  
drwxr-xr-x 2 root root 4096 May 2 2023 srv  
dr-xr-xr-x 11 root root 0 Nov 29 13:37 sys  
drwxrwxrwt 6 root root 4096 Nov 29 13:53 tmp  
drwxr-xr-x 14 root root 4096 May 2 2023 usr  
drwxr-xr-x 13 root root 4096 May 2 2023 var  
bekir@techproeducation:/$
```

Basic Shell Commands



cd [dir]	change (current) directory
cd ..	change current directory to one level up
cd /	change current directory to the root directory
cd	change current directory to the home directory

```
bekir@techproeducation: /  × + ▾ - □ ×
bekir@techproeducation:~$ pwd
/home/bekir
bekir@techproeducation:~$ cd ..
bekir@techproeducation:/home$ pwd
/home
bekir@techproeducation:/home$ cd
bekir@techproeducation:~$ pwd
/home/bekir
bekir@techproeducation:~$ cd /
bekir@techproeducation:/$ pwd
/
bekir@techproeducation:/$
```

Basic Shell Commands



mkdir [dir] create a new directory

```
bekir@techproeducation: ~  
ls  
linux-lesson  
bekir@techproeducation:~$ mkdir python  
bekir@techproeducation:~$ ls  
linux-lesson  python  
bekir@techproeducation:~$
```

Basic Shell Commands



touch create a file

```
bekir@techproeducation: ~/li x + v - □ X
bekir@techproeducation:~/linux-lesson$ pwd
/home/bekir/linux-lesson
bekir@techproeducation:~/linux-lesson$ touch file1 file2
bekir@techproeducation:~/linux-lesson$ ls
file1 file2
bekir@techproeducation:~/linux-lesson$
```



Simple Globbing

Simple Globbing



Globbing is primarily used to match patterns in filenames or text by using a wildcard character to create a pattern.

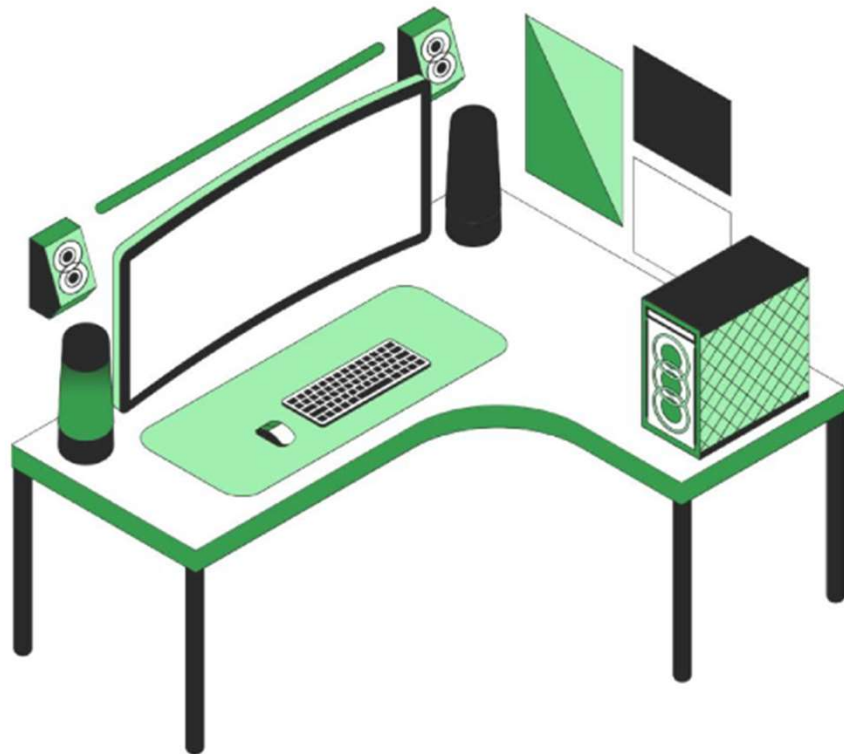
Character	Name	Function
?	Question mark	Match any single character
*	Asterisk	Match any number of character(s)
[]	Brackets	Match character from a range
{ }	Curly brace	Used to match more than one pattern

Multiple File/Directory Operations



- ? used for a single character
- * used for multiple characters.

```
bekir@techproeducation: /  
bekir@techproeducation:/$ ls  
bin  etc  lib  libx32  mnt  root  snap  tmp  
boot home lib32 lost+found opt  run  srv  usr  
dev  init lib64 media  proc sbin sys  var  
bekir@techproeducation:/$ ls lib* -l  
lrwxrwxrwx 1 root root 7 May 2 2023 lib -> usr/lib  
lrwxrwxrwx 1 root root 9 May 2 2023 lib32 -> usr/lib32  
lrwxrwxrwx 1 root root 9 May 2 2023 lib64 -> usr/lib64  
lrwxrwxrwx 1 root root 10 May 2 2023 libx32 -> usr/libx32  
bekir@techproeducation:/$ ls lib3? -l  
lrwxrwxrwx 1 root root 9 May 2 2023 lib32 -> usr/lib32  
bekir@techproeducation:/$ ls li? -l  
lrwxrwxrwx 1 root root 7 May 2 2023 lib -> usr/lib  
bekir@techproeducation:/$
```



Do you have any questions?

Send it to us! We hope you learned something new.